A Simple Research of Divisor Graphs

Yu-ping Tsao
General Education Center
China University of Technology, Taipei, Taiwan
yp-tsao@cute.edu.tw
Taipei, Taiwan 116, Republic of China

Abstract

Let S be a finite, nonempty set of positive integers. Then, the divisor graph $D(S)$ of S has S as its vertex set, and vertices i and j are adjacent if and only if either $i|j$ or $j|i$.

This paper investigates the vertex-chromatic number, the clique number, the clique cover number, and the independence number of $D([n])$ and its complement, where $[n]=\{i:1\le i\le n, n\in \mathrm{N}\}$. Besides, we discuss the perfect property on this kind of graphs. In the last section, we also give some notions about the bandwidth of the divisor graphs.

1 Introduction

In 2000 Singh and Santhosh [4] defined the concept of a divisor graph. They defined a divisor graph G as an ordered pair (V,E) where $V\subset Z$ and for all $u,v\in V$, $u\ne v$, $uv\in E$ if and only if $u|v$ or $v|u$. Singh and Santhosh [4] showed that every odd cycle of length five or more is not a divisor graph while all even cycles, complete graphs, and caterpillars are divisor graphs. In 2001, Chartrand, Muntean, Saenpholphant and Zhang [1] also studied divisor graphs. They let S be a finite, nonempty set of positive integers. Then, the divisor graph $D(S)$ of S has S as its vertex set, and vertices i and j are adjacent if and only if either $i|j$ or $j|i$. A graph G is a divisor graph if $G\cong D(S)$ for some nonempty, finite set S of positive integers. Hence, if G is a divisor graph, then there exists a function $f:V(G)\to \mathrm{N}$, called a divisor labeling of G, such that $G\cong D(f(V(G)))$. The results of [4] are confirmed in [1], where it was shown that trees and bipartite graphs are divisor graphs and a characterization of all divisor graphs was given. Moreover, Le Anh Vinh [3] showed that for any positive integer n, if m is an integer between 0 and n, then there exists a divisor graph of order n and size m. Christopher Frayer [2] gave a condition for Cartesian product of graphs such that it could be a divisor graph.

We devote the main of this article to the vertex -chromatic number, the clique number, the clique cover number, the independence number, and the bandwidth of $D([n])$, where $[n]$ means the set $\{1,2,3,\cdots,n-1,n\}$. For convenience, $D([n])$ and its complement are written simply as $D(n)$ and $\overline{D}(n)$ later, respectively.

[*] This work was supported in part by the National Science Council of the Republic of China under Contract NSC 98-2115-M-163 -001.

2 Chromatic number and clique number

The *chromatic number* of a graph G, written $\chi(G)$, is the minimum number of colors need to label the vertices so that the adjacent vertices receive different colors. The *clique number* of a graph G, written $\omega(G)$, is the maximum size of a set of pairwise adjacent vertices (clique) in G. The following lemma is trivial.

Lemma 1 For any graph G, $\omega(G)\le\chi(G)$.

Theorem 2 $\chi(D(n))=\lfloor\log_2 n\rfloor+1=\omega(D(n))$.

Proof. At first, we give an upper bound of $\chi(D(n))$ with the mapping

$$c:V(D(n))\to\{i:1\le i\le\lfloor\log_2 n\rfloor+1\}$$

defined by $c(x)=\lfloor\log_2 x\rfloor+1$. Let $c(x)=c(y)$, then $\lfloor\log_2 x\rfloor+1=\lfloor\log_2 y\rfloor+1$ implies $|\log_2 x-\log_2 y|<1$, i.e. $1/2<x/y<2$, and so $x=y$, otherwise x is not adjacent to y because x is not a divisor and a multiple of y. Since c is a proper $\lfloor\log_2 n\rfloor+1$–coloring, $\chi(D(n))\le\lfloor\log_2 n\rfloor+1$. Next, we show that $\lfloor\log_2 n\rfloor+1$ is a lower bound of $\omega(D(n))$. Clearly, $\{2^i:0\le i\le\lfloor\log_2 n\rfloor\}$ is a clique. It is easy to get $\omega(D(n))\ge\lfloor\log_2 n\rfloor+1$. From Lemma 1, we have

$$\lfloor\log_2 n\rfloor+1\le\omega(D(n))\le\chi(D(n))\le\lfloor\log_2 n\rfloor+1$$

, and the equality is obtained. ■

Theorem 3 $\chi\left(\overline{D}(n)\right)=\lceil n/2\rceil=\omega\left(\overline{D}(n)\right)$

Proof. At first, we give an upper bound of $\chi\left(\overline{D}(n)\right)$ with the mapping

$$c:V\left(\overline{D}(n)\right)\to\{i:1\le i\le\lceil n/2\rceil\}$$

defined by $c(x)=i_x$, where $x=(2i_x-1)2^{j_x}$. Let $c(x)=c(y)$, then $i_x=i_y$, and so $x=y$, otherwise x is not adjacent to y because x is a proper divisor or a proper multiple of y. Since c is a proper $\lceil n/2\rceil$ –coloring, $\chi\left(\overline{D}(n)\right)\le\lceil n/2\rceil$. Next, we show that $\lceil n/2\rceil$ is a lower bound of $\omega\left(\overline{D}(n)\right)$. For $6k-3\le n\le 6k+2$, $k\in\mathrm{N}$, let $C(n)$ be $\{i:2k\le i\le 4k-1\}\cup\{2i-1:2k+1\le i\le\lceil n/2\rceil\}$, then it is easy to check that $C(n)$ is a clique of size $\lceil n/2\rceil$. Thus $\omega\left(\overline{D}(n)\right)\ge\lceil n/2\rceil$. By Lemma 1, we have $\lceil n/2\rceil\le\omega\left(\overline{D}(n)\right)\le\chi\left(\overline{D}(n)\right)\le\lceil n/2\rceil$, and the equality is obtained. ■

3 Clique cover number and independence number

A *clique cover* of G is a partition of $V(G)$ into cliques. The minimum number of cliques in a clique cover of G is called the *clique cover number* of G and is denoted by $\theta(G)$. An *independent set* (or *stable set*) in a graph is a set of pairwise nonadjacent vertices. The *independence number* of a graph G, denoted by $\alpha(G)$, is the maximum size of an independent set of vertices. Note that the following lemmas are well-known.

Lemma 4 For any graph G, $\theta(G)=\chi\left(\overline{G}\right)$.

Lemma 5 For any graph G, $\alpha(G)=\omega\left(\overline{G}\right)$.

Theorem 6
(1) $\theta(D(n))=\lceil n/2\rceil=\alpha(D(n))$
(2) $\theta\left(\overline{D}(n)\right)=\lfloor\log_2 n\rfloor+1=\alpha\left(\overline{D}(n)\right)$

Proof.
(1) A straightforward application of Lemma 4, Lemma 5, and Theorem 3.
(2) A straightforward application of Lemma 4, Lemma 5, and Theorem 2. ■

In virtue of the corroboration of Theorem 2, it is natural that we will ask a question: Is $D(n)$ a perfect graph? We say that graph G is a perfect if $\omega(H)=\chi(H)$ for each its induced subgraph H.

Theorem 7 $D(n)$ is a perfect graph.

Proof. For $x\in\mathrm{N}$, let F be a subset of some positive factors of x such that $i|j$ or $j|i$ for $i,j\in F$. $\Gamma(x)$ denotes the set of all these F's and $\tilde{F}_x$ denotes a maximum F in $\Gamma(x)$. Let H be an induced subgraph of $D(n)$. We define a mapping c from $V(H)$ to N by

$$c(x)=\left|\tilde{F}_x\right| \text{ for } x\in V(H).$$

Let i be adjacent to j, W.L.O.G., assume $i|j$, then $\tilde{F}_i\cup\{j\}\in\Gamma(j)$ and $\left|\tilde{F}_i\cup\{j\}\right|\le\left|\tilde{F}_j\right|$. Since $j\notin\tilde{F}_i$, $c(i)=\left|\tilde{F}_i\right|<\left|\tilde{F}_i\cup\{j\}\right|\le c(j)$, and hence c is a proper coloring. Let $\rho=\max_{x\in V(H)}\left|\tilde{F}_x\right|$, then $\chi(H)\le\rho$. Let $\tilde{F}_y$ be some set having ρ elements. Because $\tilde{F}_y$ is a clique in H, $\chi(H)\le\rho\le\omega(H)$. By Lemma 1, we have $\omega(H)=\chi(H)$. This completes the proof. ■

Corollary 8 A divisor graph is a perfect graph.

Proof. In fact, a divisor graph is an induced subgraph of some $D(n)$. Since each induced subgraph of this divisor graph also is an induced subgraph of $D(n)$, with the same argument in Theorem 7, we derive the result. ■

4 Bandwidth

We call f a *numbering* of a graph G if f is a bijection from $V(G)$ to $[|V(G)|]$. When the vertices of a graph G are numbered with distinct integers, the *dilation* is the maximum difference between integers assigned to adjacent vertices. The *bandwidth* $B(G)$ of a graph G is the minimum dilation of a numbering of G. That is to say, the *dilation* of f on G, written

$Dil_f(G)$ (or $B_f(G)$), is defined as $\max\{|f(u)-f(v)|: uv \in E(G)\}$ and the *bandwidth* of G is defined as $B(G)=\min\{B_f(G): f$ is a numbering of $G\}$.

In this section, we focus the study on the bandwidth of $D(S)$ for $S \subseteq [n]$. For a start, we introduce two trivial lemmas.

Lemma 9 If H is a subgraph of G, then $B(H) \le B(G)$.

Lemma 10 If G is a complete graph of order n, then $B(G)=n-1$.

Lemma 11 If $\gcd S = \min S$, then $B(D(S)) \ge \lfloor |S|/2 \rfloor$.

Proof. Let f be a numbering of $D(S)$ and $\min S = k$. Since $\gcd S = k$, k is adjacent to each vertex in $S-\{k\}$. Therefore, we have

$$\begin{aligned} B_f(D(S)) &= \max\{|f(i)-f(j)| : ij \in E(D(S))\} \\ &\ge \max\{|f(k)-f(j)| : j \in S-\{k\}\} \\ &\ge \max\{f(k)-1, |S|-f(k)\} \\ &\ge \lfloor |S|/2 \rfloor \end{aligned}$$

Because $B_f(D(S)) \ge \lfloor |S|/2 \rfloor$ is always true for each numbering f of $D(S)$, we know that $\lfloor |S|/2 \rfloor$ is a lower bound of $B(D(S))$. ■

Theorem 12 $B(D(n)) = \lfloor n/2 \rfloor$

Proof. Since $\gcd[n]=1=\min[n]$, by Lemma 9 it is easy to see $B(D(n)) \ge \lfloor n/2 \rfloor$. Thereinafter, we show that $\lfloor n/2 \rfloor$ is also an upper bound. Let f be a function defined on $[n]$ by

$$f(i) = \frac{1-(-1)^i}{2}\left\lfloor \frac{n}{2} \right\rfloor + \left\lceil \frac{i}{2} \right\rceil.$$

After checking carefully, we can make sure that f is a numbering on $[n]$. Let E and O be the set of positive even numbers and odd numbers, respectively. And let i be adjacent to j (W.L.O .G, assume $j=ki$, $k \in \mathrm{N}$).

• If $k \in \mathrm{O}$, then

$$|f(i)-f(j)| = \left\| \left\lceil \frac{i}{2} \right\rceil - \left\lceil \frac{j}{2} \right\rceil \right\| \le \left\| \left\lceil \frac{n}{2} \right\rceil - 1 \right\| \le \left\lfloor \frac{n}{2} \right\rfloor.$$

• If $k \in \mathrm{E}$ and $i \in \mathrm{O}$, then

$$|f(i)-f(j)| = \left\| \left\lfloor \frac{n}{2} \right\rfloor + \frac{i+1}{2} - \frac{ik}{2} \right\| \le \left\lfloor \frac{n}{2} \right\rfloor.$$

• If $k \in \mathrm{E}$ and $i \in \mathrm{E}$, then

$$|f(i)-f(j)| = \left| \frac{i}{2} - \frac{ki}{2} \right| \le \left| \frac{n}{2} - 1 \right| \le \left\lfloor \frac{n}{2} \right\rfloor.$$

In a word, $|f(i)-f(j)| \le \lfloor n/2 \rfloor$ when i is adjacent to j. Thus we have $B_f(D(n)) \le \lfloor n/2 \rfloor$, and hence $B(D(n)) \le \lfloor n/2 \rfloor$. ■

Then, we study the conditions leading to $B(D(S)) = \lfloor |S|/2 \rfloor$. We call Ω a self-contained subset of S if Ω is a set contained in $S-\{1\}$ such that we have $N(x)-\{1\} \subseteq \Omega$ for each $x \in \Omega$. Let $X \subseteq S$, we use simply $N(X)$ to denote the set of vertices, outside in $S-X$, having a neighbor in X.

Theorem 13 Let $S \subset \mathrm{N}$ be a finite set containing 1. Then each of the following states will result in $B(D(S)) = \lfloor |S|/2 \rfloor$.

(1) There is a self-contained subset Ω of S such that $|\Omega| = \lfloor |S|/2 \rfloor$.

(2) There is a maximum self-contained subset Ω of S such that $|\Omega| < \lfloor |S|/2 \rfloor$.

(3) There is a self-contained subset Ω of S such that $|\Omega| > \lfloor |S|/2 \rfloor$ and there is a subset X of Ω with $|X| = |\Omega| - \lfloor |S|/2 \rfloor$ such that $|N(X)| \le 2\lfloor |S|/2 \rfloor - |\Omega|$.

Proof. Since $\gcd S = 1 = \min S$, by Lemma 7 it is clear to see $B(D(S)) \ge \lfloor |S|/2 \rfloor$. Next, we show that $\lfloor |S|/2 \rfloor$ is also an upper bound.

(1) Let f be a one-to-one function from S to $[|S|]$ such that $f(\Omega) = \{i : 1 \le i \le \lfloor |S|/2 \rfloor, i \in \mathrm{N}\}$ and $f(1) = \lfloor |S|/2 \rfloor + 1$. Let i is adjacent to j for $i, j \ne 1$, then $i, j \in \Omega$ or $i, j \notin \Omega$ by the assumption. Regardless of any case, it is easy to check that $|f(i)-f(j)| \le \lfloor |S|/2 \rfloor$. Thus we obtain $B(D(S)) \le B_f(D(S)) \le \lfloor |S|/2 \rfloor$.

(2) Let f be a one-to-one function from S to $[\![|S|]\!]$ such that $f(\Omega)=\{i:1\le i\le|\Omega|, i\in \mathrm{N}\}$ and $f(1)=\lfloor|S|/2\rfloor+1$. Let i is adjacent to j for $i,j\neq 1$, then $i,j\in\Omega$ by the assumption. It is obvious to carry off $|f(i)-f(j)|\le\lfloor|S|/2\rfloor$. Thus we obtain $B(D(S))\le B_f(D(S))\le\lfloor|S|/2\rfloor$.

(3) Let f be a one-to-one function from S to $[\![|S|]\!]$ such that

$$\begin{cases} f(\Omega - X\cup N(X)) = [\![\Omega - X\cup N(X)|]\!], \\ f(1)=\lfloor|S|/2\rfloor+1, \\ f(X\cup N(X)) = [\![|\Omega|+1]\!] - [\![\Omega - X\cup N(X)|]\!]. \end{cases}$$

Let i is adjacent to j for $i,j\neq 1$, then $i,j\in\Omega-X$, $i,j\in X\cup N(X)$, or $i,j\notin\Omega$ by the assumption. It is not difficult to check that $|f(i)-f(j)|\le\lfloor|S|/2\rfloor$. Therefore we derive $B(D(S))\le B_f(D(S))\le\lfloor|S|/2\rfloor$. ■

In the following, we give three examples to correspond with three sufficient conditions in the above theorem, respectively.

(1) Let $S=\{1,3,5,8,10,11,14,16,22,25,26,30,34,38,40,44,46,49,50\}$. Then we can find $\Omega=\{3,5,8,10,16,25,30,40,50\}$ of order $9=\lfloor 19/2\rfloor$. And hence the numbering f of $D(S)$, defined in the table below, may prove $B(D(S))=9$.

i	1	3	5	8	10	11	14	16	22	25
$f(i)$	10	1	2	3	4	11	12	5	13	6
i	26	30	34	38	40	44	46	49	50	
$f(i)$	14	7	15	16	8	17	18	19	9	

(2) Let $S=\{1,4,5,8,10,11,16,18,22,25,26,30,34,38,40,41,42,49,51\}$. Then we can find a maximum $\Omega=\{4,5,8,10,16,25,30,40\}$ of order $8<\lfloor|S|/2\rfloor=\lfloor 19/2\rfloor=9$. And hence the numbering f of $D(S)$, defined in the table below, may prove $B(D(S))=9$.

i	1	4	5	8	10	11	16	18	22	25
$f(i)$	10	1	2	3	4	9	5	11	12	6
i	26	30	34	38	40	41	42	49	51	
$f(i)$	13	7	14	15	8	16	17	18	19	

(3) Let $S=\{1,5,8,10,12,14,16,17,22,24,25,26,32,34,36,40,42,44,51\}$. Then we can find a $\Omega=\{5,8,10,14,16,22,25,32,40,42,44\}$ of order $11>\lfloor|S|/2\rfloor=\lfloor 19/2\rfloor=9$ and a $X=\{42,44\}$ with $N(X)=\phi$, where

$$\begin{cases} |X|=2=|\Omega|-\lfloor|S|/2\rfloor, \\ |N(X)|=0\le 7=2\lfloor|S|/2\rfloor-|\Omega|. \end{cases}$$

And hence the numbering f of $D(S)$, defined in the table below, may prove $B(D(S))=9$.

i	1	5	8	10	12	14	16	17	22	24
$f(i)$	10	1	2	3	13	4	5	14	6	15
i	25	26	32	34	36	40	42	44	51	
$f(i)$	7	16	8	17	18	9	11	12	19	

For $n,d\in\mathrm{N}$, let

$$A(n,d)=\{a_i : a_i=1+(i-1)d, i\in[n]\}.$$

We attempt to testify $B(D(A(n,d)))=\lfloor n/2\rfloor$ by applying Theorem 13.

Corollary 14 $B(D(A(n,d)))=\lfloor n/2\rfloor$ for $d\in\mathrm{N}$.

From the results of Theorem 10 and Theorem 11, of course we will surmise the following conjecture is right:

Let $S\subset\mathrm{N}$ be a finite set containing 1, then $B(D(S))=\lfloor|S|/2\rfloor$.

We found a counterexample easily as follows：
Let $S=\{1,2,3,4,5,7,8,9,16,32,64\}$. Since $\{1,2,4,8,16,32,64\}\subset S$ is a clique of order 7 in $D(S)$, by Lemma 9 and Lemma 10, we get

$$B(D(S))\ge 7-1=6>5=\lfloor 11/2\rfloor=\lfloor|S|/2\rfloor.$$

In the finality for bandwidth of divisor graphs, we give an easy outcome to clarify the wrong guess.

Proposition 15 Given any $m\in\mathrm{N}-\{1\}$, for $\lfloor m/2\rfloor\le k\le m-1$, there is a finite subset S of N containing 1 such that $|S|=m$ and $B(D(S))=k$.

Proof. Let $A = \{2^i : i \leq k, i \in \mathrm{N} \cup \{0\}\}$, and B be a set of $m-k-1$ odd numbers. Then it could be verified that $A \cup B$ is the set S what we need. First at all, since A is a clique of order $k+1$, we get $B(D(S)) \geq |A| - 1 = k$ from Lemma 9 and Lemma 10. Then we look for an upper bound. Let f be a one-to-one function from S to $[|S|]$ such that

$$\begin{cases} f(A - \{1\}) = [k-1], \\ f(1) = k+1. \end{cases}$$

Let i is adjacent to j for $i, j \neq 1$, then $i, j \in A$ or $i, j \in B$ by the assumption. Regardless of any case, it is easy to check that $|f(i) - f(j)| \leq k$. Consequently, we obtain $B(D(S)) \leq k$. And the equality is acquired. ■

5 Conclusion and future works

Divisor graphs are special and look like easily to begin. This work has provided some results on divisor graphs. We think that it will be interesting for exploring the other parameters of $D(n)$, such as the maximum size of matching, arboricity, profile, circumference, edge-chromatic number, acyclic chromatic number, acyclic chromatic index, etc.

References

[1] Western G. Chartrand, R. Muntean, V. Saenpholphat, and P. Zhang, Which Graphs Are Divisor Graphs ? Congr. Numer. 151(2001) 189-200.

[2] Christopher Frayer, Properties of Divisor Graphs.

[3] Le Anh Vinh, Divisor graphs have arbitrary order and size, AWOCA 2006.

[4] G. S. Singh and G. Santhosh, Divisor Graphs-I. Preprint (2000)